

UPDATED ALLSERVE 4-WEEK PROJECT PLAN (FINAL VERSION)

WEEK 1: FOUNDATION, ARCHITECTURE & AUTH SYSTEM (UPDATED)

Modules Covered

- Authentication (User / Provider / Admin)
 - User Profile + Address + Payment Methods
 - Provider Application System
 - Category + Service System
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1. PROJECT FOUNDATION

- Node.js + Express + TypeScript setup
- Modular Monolith + Layered Architecture

```
modules/  
├── auth/  
├── user/  
├── provider/  
└── services/
```

- Repository pattern (optional but structured)
 - Environment config + logger (Winston)
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2. AUTH SYSTEM

User Auth

- Email/password signup

- Google login
- Email OTP verification
- Phone number + OTP verification
- JWT login

Provider Auth (NEW FLOW)

- Apply as provider (separate)
- Upload documents
- Application status tracking
- Provider login only after approval

Admin Auth

- Admin login
 - Role-based access control
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3. USER MODULE Profile APIs

- Upload profile image
 - Address system (multiple addresses)
 - Payment methods (UPI / card)
 - Favorites system
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4. PROVIDER APPLICATION SYSTEM

- Apply → Upload docs → Verify → Approve/Reject
 - Separate collection (**ProviderApplications**)
 - Admin verification flow
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5. SERVICES SYSTEM

- Provider creates services (catalogue)
- Each service:
 - name, price, duration, images
- Link:

Provider → Services → Category

6. CATEGORY + DISCOVERY

- Category CRUD
- Service-based filtering
- Basic search (location + category)

7. SECURITY + API STANDARDS

- Zod/Joi validation
- JWT middleware
- Role-based access
- Global error handler
- Standard response format
- Rate limiting + CORS + Helmet

WEEK 1 OUTCOME

Fully working auth system (OTP + roles)
User + Provider onboarding complete
Service-based marketplace structure ready

WEEK 2: AVAILABILITY, BOOKING & REQUEST SYSTEM (CORE ENGINE)

1. SLOT / AVAILABILITY SYSTEM

- Provider creates slots
- Prevent overlapping slots
- Slot locking during booking

APIs:

/provider/slots
/providers/:id/slots



2. BOOKING SYSTEM

Schema additions:

- serviceId
- addressId
- paymentId

Flow:

Create → Accept → Complete / Cancel

Features:

- Booking details page
 - Update booking (date/address)
 - Provider accept/reject
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3. REQUEST SYSTEM

New features:

- Budget field
- Request tracking
- Provider responses

New collection:

RequestResponses

Flow:

User → Request → Providers respond → User selects → Booking



4. BASIC INTERACTION

- Request-level messaging (before full chat)
 - Provider quotes
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WEEK 2 OUTCOME

Complete booking engine

Job-based marketplace system (request flow)

WEEK 3: PAYMENT, CHAT, REVIEWS & NOTIFICATIONS (MAJOR UPGRADE)

1. PAYMENT SYSTEM

- Razorpay integration
- Create order + verify payment
- Booking confirmed after payment

Add:

- invoiceUrl
 - payment status tracking
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2. INVOICE SYSTEM

- Generate invoice
 - Download PDF
 - Link to booking
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3. REFUND SYSTEM

- Refund on cancellation
 - Update payment + booking
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4. CHAT SYSTEM

New collections:

Conversations

Messages

Features:

- Chat with provider (sidebar UI)
 - Message history
 - Read/unread tracking
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★ 5. REVIEWS

- Add review after completed booking
 - Rating aggregation
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🔔 6. NOTIFICATIONS

Add:

- type (booking, payment, message)
- isRead

Triggers:

- Booking updates
 - Payment updates
 - Chat messages
 - Request responses
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🧠 WEEK 3 OUTCOME

Real-time-like system (chat + notifications)

Complete payment + invoice flow

Trust system (reviews)

🔴 WEEK 4: ADMIN, ANALYTICS, AI & SYSTEM COMPLETION

👤 1. ADMIN PANEL

Features:

- User management (block/unblock)
 - Provider management
 - Provider application approval system
 - Booking control
 - Payment export (CSV)
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2. ANALYTICS

- Dashboard stats
 - Revenue tracking
 - Provider performance
 - Category insights
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3. AI SYSTEM

- Smart search (location + query)
 - Recommendation logic:

rating + price + availability + response rate
 - Price estimation
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4. SUPPORT SYSTEM (ENHANCED)

- Ticket system (user + provider)
 - Admin responses
 - Ticket status tracking
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5. SETTINGS

- Notification preferences
 - Language selection
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6. FINAL SYSTEM HARDENING

- Input validation everywhere
 - Secure endpoints
 - Optimize DB queries
 - Add indexes
 - Clean code (SOLID)
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7. TESTING

- End-to-end testing:

Auth → Booking → Payment → Review
 - Edge cases
 - Error handling
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8. DOCUMENTATION

- Postman (complete)
 - API docs
 - Architecture explanation
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WEEK 4 OUTCOME

Fully complete marketplace system
Admin control + analytics + AI
Production-ready backend structure